

Solving equations using evidence and zero.

So, we have (as an example)

$$x^4 + 3x^3 + 5x^2 + x = 2$$

Subtract 2 by both sides

$$x^4 + 3x^3 + 5x^2 + x - 2 = 0$$

Put x in evidence.

$$x(x^3 + 3x^2 + 5x + 1 - \frac{2}{x}) = 0$$

$$x^3 + 3x^2 + 5x + 1 - \frac{2}{x} = \frac{0}{x}$$

So, dividing zero, by x, right side

$$x^3 + 3x^2 + 5x + 1 - \frac{2}{x} = 0$$

putting x in evidence again

$$x(x^2 + 3x + 5 + \frac{1}{x} - \frac{2}{x^2}) = 0$$

$$x^2 + 3x + 5 + \frac{1}{x} - \frac{2}{x^2} = \frac{0}{x}$$

dividing zero by x, on right side.

Equation 1

$$x^2 + 3x + 5 + \frac{1}{x} - \frac{2}{x^2} = 0$$

End Equation 1

In this, case , we have a parabola formula (On equation 1)

Also,

$x^4 + 3x^3 + 5x^2 + x = 2$ is a parabola formula because of equation 1

To consider

On, a equation, we have A=B...what if we just develop A, and left B alone?

$$x^5 + 2x^4 + 6x^2 = x+1$$

keep developing left side, and let right side the same

Example

$$A^{x-1} * C - A^x = B^y \quad (1)$$

$$A^x \left(\frac{1}{A} * C - 1 \right) = B^y \quad (2)$$

$$A^x = \frac{B^y}{\frac{C}{A} - 1} \quad (3)$$

$$A^x = \frac{B^y}{\frac{C-1}{A}} \quad (4)$$

$$A^x * A = \frac{B^y}{C-1} \quad (5)$$

$$A^x * A * (C-1) = B^y \quad (6)$$

So, we can solve the equation by solving 1,2 and 6...no need extra arithmetic. (steps 3,4 and 5).

